Where Occupations Move:
Interstate Migration by Occupation in the United States, 2005 to 2024

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Extended summary

Interstate migration in the United States is well documented for people and barely documented for occupations. This matters because most of the rules that make moving difficult attach to jobs rather than to places, and because a state deciding whether to train workers or to recruit them needs to know how much of its workforce arrives from outside, which occupations that applies to, and who those arrivals are. A national migration rate answers none of those questions.

This paper builds bilateral interstate migration flows by occupation from nineteen vintages of American Community Survey microdata covering 2005 to 2024, 59.8 million person records across 51 jurisdictions and 479 harmonised occupation codes, and uses them to construct a measure of how much of each state's occupational workforce arrives from another state within the year. The measure ranges thirteenfold across occupations, from 0.66 per cent for tool and die makers to 9.70 per cent for sailors, with design-based confidence intervals that do not overlap between the two ends of the distribution.

Three findings follow. First, interstate migration delivers young workers and delivers them in both directions: arrivals average 34.0 years of age, leavers average 34.0, and the workforce they join or leave averages 41.3. Across every occupation with enough arrivals to measure, incomers are younger than the people already doing the job, so no occupation imports experience and a state short of experienced workers cannot recruit its way out of the shortage. Second, a gravity model estimated by Poisson pseudo-maximum likelihood puts the elasticity of flows with respect to the origin-destination wage differential at 0.114, against a distance elasticity of roughly -0.69, so a ten per cent pay advantage buys about one per cent more migration while distance matters some six times more. Third, deviations from the gravity benchmark recover known occupational agglomerations without being told about them, placing Washington state at 2.82 times its predicted intake of software developers, the District of Columbia at 3.64 times its predicted intake of lawyers, and Michigan at 4.16 times its predicted intake of mechanical engineers.

The measures are tested against the Occupational Employment and Wage Statistics, an establishment survey independent of the household microdata from which every migration figure here is built. Ten hypotheses were specified in advance and all ten are reported. Three hold after correction for multiple testing: the gravity residual tracks the location quotient, external dependence predicts employment growth, and net inflow predicts employment growth. The wage tests fail, and they fail precisely rather than for want of power, with coefficients of -0.0004, -0.0014 and $+0.0001$ against standard errors between 0.004 and 0.007. Two unrelated data sources therefore agree that

interstate migration tracks where employment is expanding and bears no detectable relationship to where pay is rising.

1 Introduction

Occupation is the unit at which mobility differs

Roughly 2.3 per cent of employed Americans crossed a state line in a given year over the period studied here, a figure stable enough that it conceals almost everything of interest. Within it, tool and die makers replenish two thirds of one per cent of their number from other states each year while sailors replenish close to a tenth. Farmers, postal workers, bus drivers and highway maintenance workers sit within a percentage point of the floor; pilots, air traffic controllers, aircraft mechanics and physical scientists sit five to nine times above it. Any policy instrument aimed at labour mobility acts on this distribution rather than on its mean, and the distribution has not previously been measured at occupation detail across a panel of this length.

The gap reflects what the available data could support. The migration literature has treated occupation coarsely or not at all, from Molloy et al. (2011) on the aggregate decline to Kennan and Walker (2011) on the individual decision, while the literature that does resolve occupation, chiefly the studies of individual licensure compacts, treats one occupation at a time. Between the two sits the object this paper constructs: bilateral flows by occupation, state and year, over a period long enough to observe change.

What a state actually needs to know

The practical question a workforce authority asks is narrower than the academic one. It is whether the people filling a given occupation locally were trained locally or arrived from elsewhere, whether the ones arriving are the ones needed, and whether recruitment or training is the instrument more likely to work. Each of those is answerable from bilateral flows resolved by occupation and age, and none is answerable from an aggregate rate.

This paper is descriptive in its claims and does not identify a causal effect of any policy. What it offers instead is a set of measures constructed from public data, reported with the uncertainty the survey design implies, and validated against a source from which they were not built.

Structure

Section 2 describes the data and the construction of the flow panel. Section 3 reports the scale of external dependence across occupations. Section 4 turns to who moves, and to the finding that migration supplies young workers to every occupation and experienced workers to none. Section 5 estimates the gravity model and reports the two frictions that govern flows. Section 6 uses departures from the gravity benchmark to locate occupational agglomeration. Section 7 tests the measures against outside data, and Section 8 sets out the limitations.

2 Data and construction

The panel

The flows are built from IPUMS USA extracts of the American Community Survey one-year files for 2005 to 2024 (Ruggles et al., 2024), 59.8 million person records in total. There is no 2020 vintage because the Census Bureau suspended one-year collection that year, so the panel runs nineteen vintages with a gap, and any calculation crossing 2019 to 2021 measures two calendar years rather than one. Occupation is coded on the OCC2010 basis, the IPUMS harmonisation, which yields 479 occupations present across the panel and avoids the step changes at the 2010 and 2018 revision boundaries that a crosswalk to the contemporaneous coding introduces.

A worker is an interstate mover if the state of residence one year previously is a valid state other than the current one. This rule is a strict subset of the survey's own interstate migration flag: no record is classified as a mover by the rule and not by the flag, while 0.7 per cent of movers carry the flag without a recorded origin state and are therefore excluded, since a flow cannot be assigned to a pair without one.

Flows with the zeros retained

Flows are counted for every origin, destination, occupation and year, and the zeros are kept. Of the cells in the estimation grid, 91.6 per cent record no movers, and a corridor carrying nobody is evidence about the difficulty of that move rather than missing data. Dropping zeros, which taking logarithms of flows requires, would estimate the model on a selected sample of large corridors and would bias the result in a direction that cannot be signed in advance.

Wages measured from workers who did not move

The wage panel gives the mean log hourly wage by occupation, state and year, built from wage and salary workers who did not cross a state line, a restriction adopted because, were the movers' own wages included, the destination wage would absorb the selection the model exists to detect, because workers who move in response to a good offer earn more on arrival, and the estimated elasticity would then be partly a construction of the measurement. Movers are 2.3 per cent of the workforce, so the cost of excluding them is small.

Hourly rather than annual pay is used, computed from annual wage income, usual weekly hours and weeks worked, which changes the concept rather than the precision. Reliability, defined as the share of observed variation that is not sampling noise, is 0.762 for cross-sectional wage differences under either measure, so the hourly figures are no more precise, but they remove an arbitrary threshold on hours that the annual measure required and they are the price a worker actually faces.

Geography and uncertainty

Distance is the great-circle distance between state centres of population rather than geographic centroids, so that Nevada is measured from the vicinity of Las Vegas rather than from empty desert, and contiguity is derived from the Census county adjacency file rather than transcribed. Standard

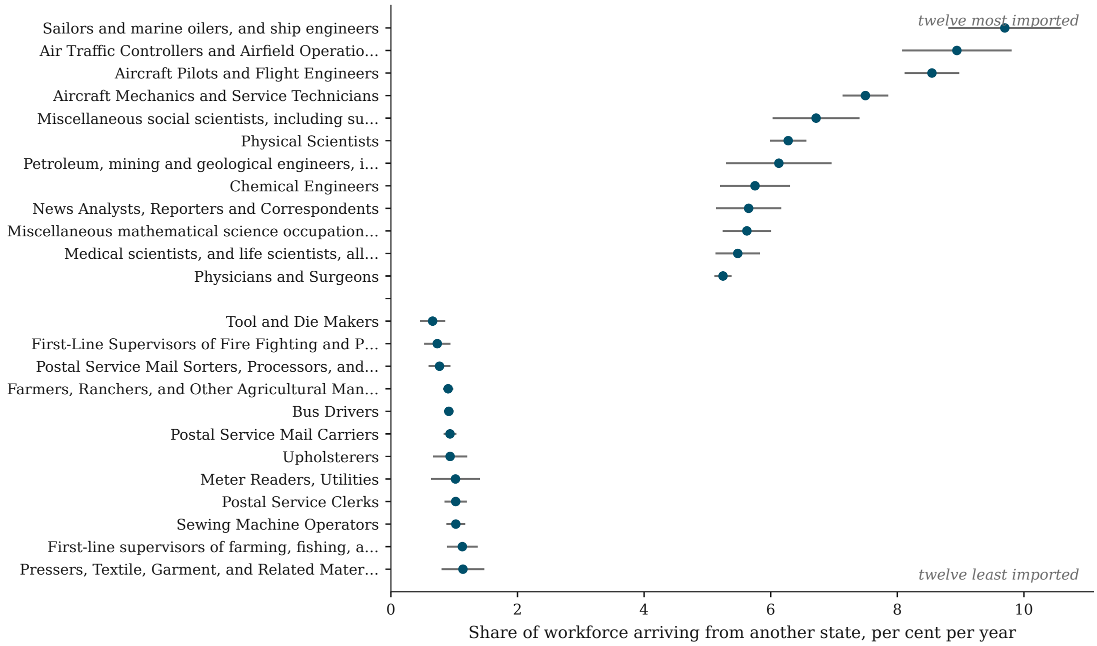

Figure 1: External dependence by occupation, pooled 2005 to 2024. Points are the share of the employed workforce arriving from another state within the year; bars are 95 per cent intervals from Taylor linearisation on the survey design. Occupations with fewer than 5,000 person records are omitted.

errors on the sourcing measures are computed by Taylor linearisation on the survey's strata and clusters, which the ACS design supports without collapsing: 2,462 strata in the final year, a minimum of 191 clusters per stratum and no singletons. The median relative standard error across occupations is 6.1 per cent, and 345 of 475 occupations fall below 10 per cent.

3 The scale of external dependence

External dependence is the share of a state's employed workforce in an occupation that arrived from another state within the year. Pooled across states and vintages it ranges from 0.66 per cent for tool and die makers to 9.70 per cent for sailors and marine oilers, a thirteenfold spread whose ends are separated by an order of magnitude even at the bounds of their confidence intervals, with the most imported occupation bounded below at 8.80 per cent and the least bounded above at 0.86. Figure 1 reports the two ends of the distribution.

Occupations at the top of the distribution either serve networks that span states by construction, as sailors, pilots, air traffic controllers and aircraft mechanics do, or draw on a national market for scarce training, as physical scientists, chemical engineers and physicians do. Occupations at the bottom are attached to a fixed local asset: a farm, a postal route, a stretch of highway, a municipal fire service. Licensure does not organise the distribution. Physicians and veterinarians are among the most heavily licensed occupations in the country and sit near the top of the ranking, while postal workers require no licence and sit near the bottom.

Gross movement is roughly twice net movement. Nationally, arrivals plus departures run at 4.5 to 5.4 per cent of employment against an arrival rate near 2.3 per cent, and the gap between the

Figure 2: Left: annual external dependence by age band, all occupations pooled. Right: mean age of arrivals against mean age of the existing workforce, one point per state and occupation with at least 100 arrivals and 100 departures. Points below the diagonal indicate arrivals younger than the workforce they join.

two widens at state level: the District of Columbia replaces close to a fifth of its workforce across state lines each year while its net balance is approximately zero. Reporting net flows alone would describe that as an absence of activity.

4 Migration and the life cycle

That migration is concentrated among the young is long established (Rogers and Castro, 1981). What the occupational resolution adds is that the pattern admits no exceptions and that the flow is symmetric in age.

Annual external dependence falls monotonically with age, from 4.54 per cent among workers under 25 to 0.98 per cent among those aged 55 and over, a fourfold-and-a-half gradient shown in the left panel of Figure 2. The mean age of arrivals is 34.0 years, the mean age of leavers is 34.0, and the mean age of the workforce they join or leave is 41.3. That arrivals and leavers should be identical in age is not an averaging of opposing tendencies: the median gap between them across state and occupation cells is -0.2 years and two thirds of cells fall within two years of each other.

Every point in the right panel of Figure 2 lies below the diagonal. Across all occupations with at least 2,000 observed arrivals, incomers are younger than the workforce they join, from 2.9 years younger for waiters to 11.9 years for aircraft mechanics, 11.0 for physicians and 10.8 for postsecondary teachers. Interstate migration therefore neither ages nor rejuvenates a state's workforce in any occupation. It replaces its early-career members with other early-career members drawn from elsewhere.

A state facing a shortage of experienced workers in an occupation cannot recruit its way out of it through interstate migration, because the channel does not carry experienced workers in any

Table 1: Wage elasticity of interstate migration flows, Poisson pseudo-maximum likelihood, standard errors clustered by state pair.

Fixed effects	Elasticity	Std. error	p
Origin, destination, year	0.075	0.022	0.0006
Origin-year, destination-year, occupation	0.109	0.026	<0.0001
+ state pair	0.108	0.026	<0.0001
+ occupation-year	**0.114**	0.026	<0.0001

occupation at any level of pay. Training and early-career retention are the instruments that remain, and they operate on the same cohort that recruitment competes for.

5 What moves workers

Specification

Flows are estimated by Poisson pseudo-maximum likelihood, which handles the zeros and the heteroskedasticity that a log-linear specification mishandles (Santos Silva and Tenreyro, 2006), on 4.02 million origin by destination by occupation by year cells. Estimating one occupation at a time is not viable, and the attempt returns a clean null that is an artefact of the design. Within a single occupation, the wage differential is identified only from movement in a state by occupation cell mean over time, and a reliability calculation attributes 54 per cent of that movement to sampling noise, so the coefficient is attenuated to roughly half its value against standard errors that cannot detect it. Pooling occupations identifies the elasticity from variation across occupations within a state pair and year, whose reliability is 0.762, and it permits origin-by-year and destination-by-year effects that absorb the time-varying multilateral resistance which would otherwise sit in the error term (Anderson and van Wincoop, 2003; Bertoli and Fernández-Huertas Moraga, 2013; Head and Mayer, 2014).

Two frictions of very different size

Table 1 reports the wage elasticity across four specifications of increasing severity. It is stable, and it rises as effects are added, which indicates that the looser specifications were biased towards zero by omitted multilateral resistance rather than the tighter ones biased upward. The preferred estimate survives origin-by-year, destination-by-year, occupation-by-year and dyad effects together, so it is identified from how wage differentials vary across occupations within one state pair in one year.

An elasticity of 0.114 means that a destination paying ten per cent more than the origin draws about one per cent more migration in that occupation. Inverting it, a state seeking ten per cent more arrivals through pay alone would have to raise the occupation's wage by something approaching ninety per cent. Correcting for the measurement reliability of the wage variable raises the estimate to approximately 0.15 and does not change that conclusion.

The distance elasticity is an order of magnitude larger in its effect and, unlike the wage elasticity, it can be estimated occupation by occupation, because it is identified from variation across 2,550 state pairs rather than from movement over time. It is statistically distinguishable from zero in

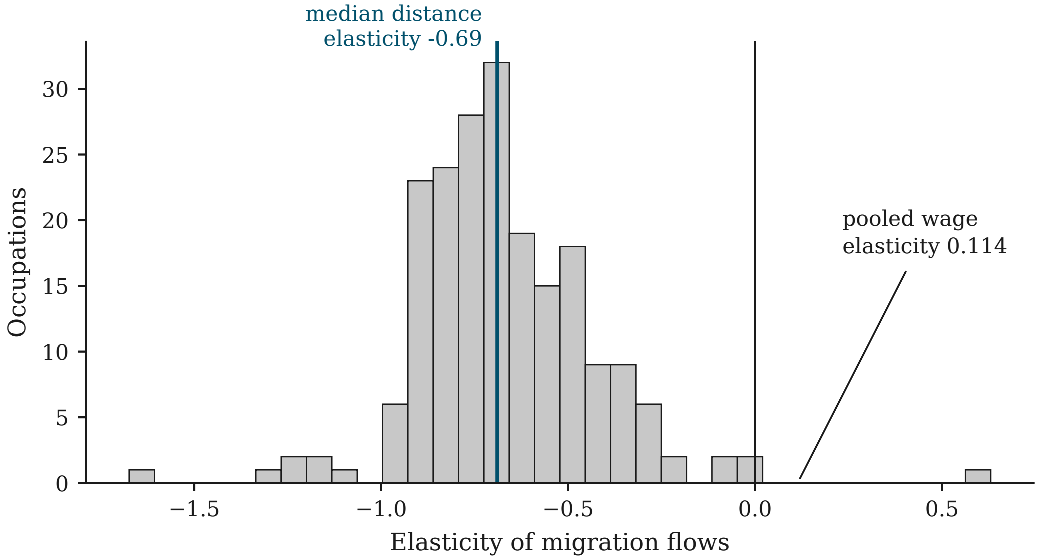

Figure 3: Distribution of the distance elasticity across 203 occupations estimated separately, with the pooled wage elasticity marked for scale.

93 per cent of the 203 occupations with sufficient corridors, and it varies threefold across them, from -0.04 for editors and -0.06 for chemists to -1.21 for printing press operators and -1.31 for roofers. Figure 3 places the pooled wage elasticity against that distribution. The interpretation is direct: a coefficient near zero describes an occupation recruited nationally, and one near unity describes an occupation for which recruitment beyond neighbouring states is largely wasted.

Movement without accumulation

Occupations differ twentyfold in the share of their workforce crossing a state line each year in either direction, from 1.29 per cent for tool and die makers to 25.52 per cent for avionics technicians. Almost none of that traffic accumulates. Aircraft pilots move 17.09 per cent of their workforce across state lines annually for a net balance of 0.000 per cent, and aircraft mechanics move 14.98 per cent for 0.011. Figure 4 shows the pattern across all occupations, with net balance confined to a narrow band around zero irrespective of how much movement there is.

The distinction bears on which instrument a state can use. In a high-turnover occupation every worker it attracts is matched by one it loses, so retention rather than recruitment is the binding constraint and additional recruitment replaces the leak without raising the stock. In a low-turnover occupation the channel is too narrow to recruit through at all, and the workforce has to be produced locally.

6 Departures from the benchmark

Once the model accounts for distance, contiguity, wage differentials and everything constant about each state in each year, what remains in a state by occupation cell is the part that those forces do not explain. Figure 5 plots observed against predicted arrivals.

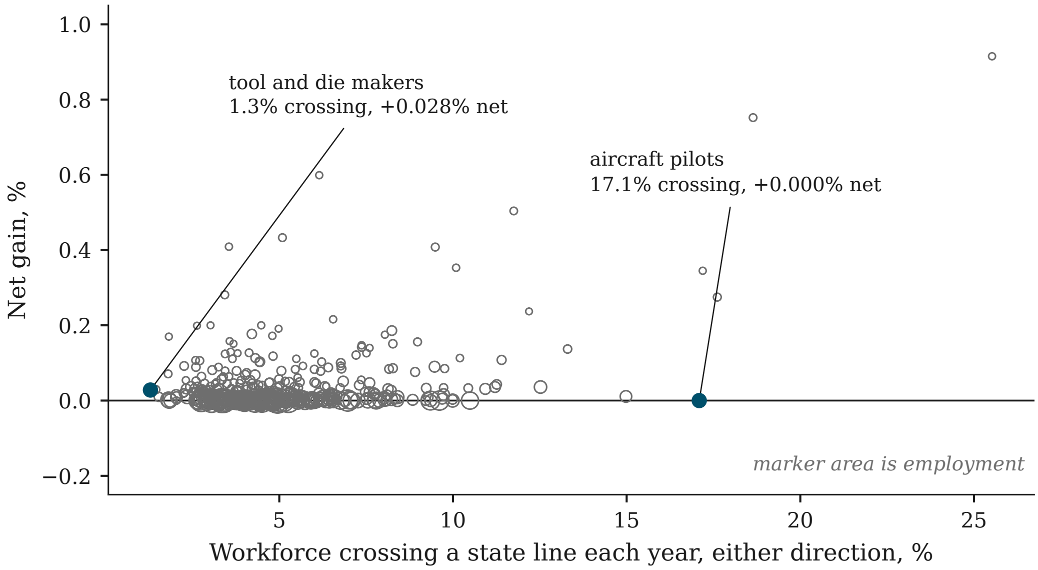

Figure 4: Annual crossing rate against net balance, one point per occupation, marker area proportional to employment.

The residual recovers occupational concentrations that were not supplied to the model. Washington state draws 2.82 times its predicted intake of software developers and California 2.03, the District of Columbia 3.64 times its predicted intake of lawyers, Michigan 4.16 times its predicted intake of mechanical engineers, and New York 2.76 times its predicted intake of securities and financial services agents. Under-supply is as legible, with South Carolina, Tennessee and Kentucky drawing between 0.28 and 0.44 of their predicted intake of software developers. California draws twice its predicted software developers and half its predicted home health aides, a pattern consistent with the cost of living sorting arrivals by what they earn.

Deviations are judged against Poisson sampling variation, since ranking on the raw ratio surfaces only small-cell artefacts: the standard deviation of the ratio falls from 0.597 in cells below 2,000 expected arrivals to 0.203 above 50,000. On the noise-adjusted measure, 43 per cent of cells exceed a z of 2 against the 5 per cent that sampling variation alone would produce.

These concentrations are also stable over the period. Aggregating the residual into four multi-year windows on a balanced panel, the correlation between the 2005 to 2009 and 2021 to 2024 positions is 0.543, and the noise-corrected spread of the residual is flat across the four windows at 0.263, 0.276, 0.257 and 0.275. Across twenty years containing a financial crisis, a pandemic and a large and sudden expansion of remote work, the geography of occupational concentration does not measurably move.

7 Testing the measures against outside data

Every measure above is built from household survey records. The Occupational Employment and Wage Statistics is collected from employers instead, so agreement between the two is informative rather than circular. Ten hypotheses were specified before the results were examined, all ten are reported, and p values carry a Benjamini-Hochberg correction across the battery (Benjamini and

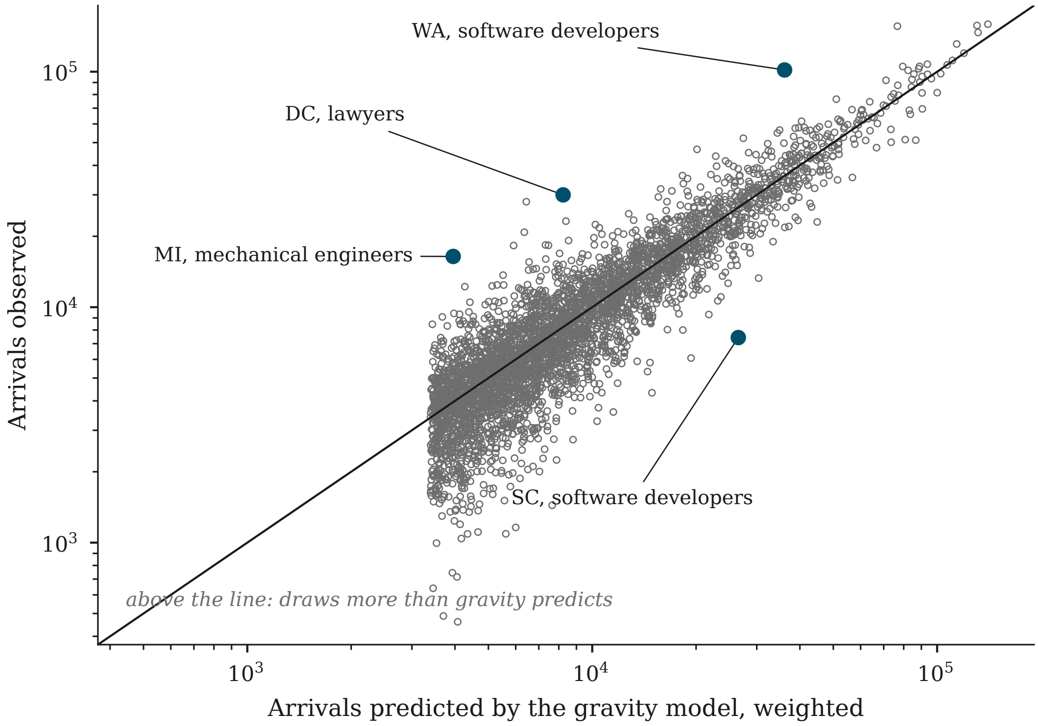

Figure 5: Observed against gravity-predicted arrivals by state and occupation, pooled 2005 to 2024, log scales. Cells below roughly thirty expected sampled movers are suppressed.

Hochberg, 1995). Regressions are weighted by employment and clustered by occupation and by state.

Three hold. The gravity residual tracks the location quotient with a standardised coefficient of 0.134 and an adjusted p of 4.4×10^{-7}, which is the result reported in Figure 6 and which establishes that the residual measures occupational concentration rather than noise. External dependence predicts subsequent employment growth at 0.044, and net inflow predicts it at 0.029.

The wage tests fail. Whether the predictor is the gravity residual, external dependence or net inflow, the relationship with five-year wage growth is -0.0004, -0.0014 and $+0.0001$ respectively, against standard errors between 0.004 and 0.007. These are measured zeros rather than imprecise estimates, and they carry a direct consequence: a state drawing materially fewer workers than the benchmark predicts experiences no faster wage growth, so the residual cannot be interpreted as a shortage signal and is not presented as one here. A tenth result points the same way, with states paying above the national rate for an occupation importing very slightly fewer of them, wrong-signed and not significant after correction, which is what a cost of living effect would look like.

The pattern across the battery is internally consistent with the gravity estimate. The model puts the wage elasticity at 0.114, and an independent employer survey agrees that interstate migration tracks where employment is expanding while bearing no detectable relationship to where pay is rising.

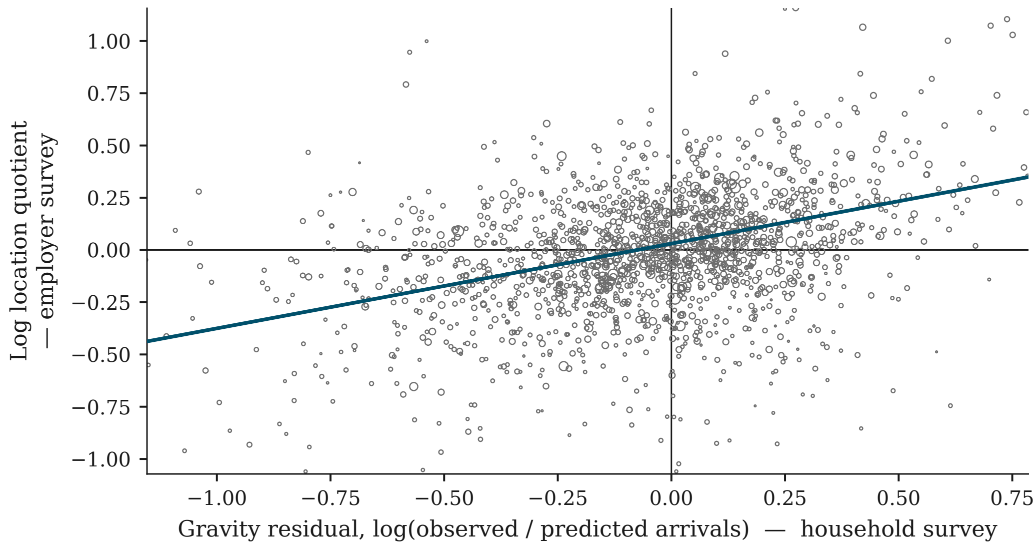

Figure 6: The gravity residual, from household survey microdata, against the location quotient, from the establishment survey. Marker area proportional to employment; the line is the employment-weighted fit.

8 Limitations

Occupation and age are recorded at the time of survey, after any move has taken place. An arrival counted in an occupation either practised it before moving or entered it on arrival, and the data cannot separate the two. For occupations with long training requirements the ambiguity is minor, while for easily entered occupations the sourcing measure is an upper bound on imported trained workers, and the policy claims are restricted accordingly.

Coverage is uneven. The results are secure nationally, by occupation, and for large combinations of state and occupation, but only 4,026 of 24,102 state by occupation cells reach thirty arrivals and thirty departures even pooling all nineteen vintages, so a complete lookup across every state and every occupation is not available from these data and is not offered. Cells below the threshold are suppressed rather than interpolated. The absent 2020 vintage makes any calculation spanning 2019 to 2021 a two-year change.

Nothing estimated here identifies a causal effect. The residual locates where a state diverges from its benchmark and is silent on the reason, and the association between migration and employment growth runs in a direction these data cannot determine.

9 Conclusion

Interstate migration supplies a thirteenfold range of occupational workforces with a channel that is narrow in every case and narrower in some than others, and that carries workers of one age. The measures constructed here put numbers on both facts, report them with the uncertainty the survey design implies, and agree with an independent employer survey where agreement is testable.

For a state deciding how to staff an occupation, three results bear directly. Pay is a weak instrument, since a ten per cent advantage buys about one per cent more arrivals and the wage tests against outside data return nothing. Recruitment cannot supply experience, since no occupation draws arrivals older than the workforce they join. And in the occupations where movement is heaviest, arrivals and departures cancel, so the constraint is retention rather than attraction.

The apparatus assembled here supports a question it does not answer. Deviations from the gravity benchmark are large, stable over two decades and unexplained, and locating them is a different exercise from accounting for them. Occupational licensure compacts, which take effect between specific pairs of states on specific dates, provide variation of the kind that could account for part of them, and that is the direction the work takes next.

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